

A Dirac-mass counterexample to the printed Beurling ultradistribution criterion on $SU(2)$

A full negative answer and corrected theorem for Conjecture 4.5.5
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Status. `candidate_full_counterexample_likely_valid_needs_human_review.`

Abstract

The second clause of Conjecture 4.5.5 in the source characterizes the dual of a sub-Laplacian Gevrey–Beurling class on $SU(2)$ using a *positive* exponential Fourier weight. It is false. The Dirac mass at the identity is an ordinary distribution and hence belongs to that dual, as the source itself records, but its Fourier blocks are identity matrices. The proposed norms grow at least as $\sqrt{2}e^{Bl^{1/(2s)}}$. We also prove the exact repair: the exponent in the second clause must be negative. With that correction, the standard weighted Hilbert-scale argument proves both clauses.

1 The printed conjecture

Let X, Y be the canonical Hörmander pair on $SU(2)$ and $\mathcal{L} = -(X^2 + Y^2)$. Irreducible representations are indexed by $l \in \frac{1}{2}\mathbb{N}_0$, have dimension $d_l = 2l + 1$, and the source gives

$$\widehat{\mathcal{L}}(l)_{m,n} = (l(l+1) - m^2)\delta_{m,n}, \quad -l \leq m, n \leq l. \quad (1)$$

For $1 \leq s < \infty$, put $a = 1/(2s)$. The first clause of source Conjecture 4.5.5 asserts that the Roumieu dual consists exactly of those u for which, for every $B > 0$,

$$\sup_l \|e^{-B\widehat{\mathcal{L}}(l)^a} \widehat{u}(l)\|_{\text{HS}} < \infty. \quad (2)$$

The second clause asserts that v belongs to the Beurling dual exactly when, for some $B > 0$,

$$\sup_l \|e^{+B\widehat{\mathcal{L}}(l)^a} \widehat{v}(l)\|_{\text{HS}} < \infty. \quad (3)$$

The positive sign in (3) is the issue. The source PDF included with this packet contains the definitions, the inclusion of ordinary distributions, and the conjecture on pages 147–149.

2 Proof intuition

A dual of a Gevrey test-function space must allow Fourier growth; a positive exponential instead demands Fourier decay. The simplest diagnostic is the Dirac mass. It is unquestionably in the

target dual, while every one of its Fourier blocks is the identity. The least sub-Laplacian eigenvalue in the spin- l block already tends to infinity, so even the two extremal diagonal entries make the proposed norm explode.

For the repair, weighted L^2 domains of $e^{D\mathcal{L}^a}$ form a Hilbert scale. Taking the dual reverses the weight, and exchanging Roumieu “some D ” with Beurling “every D ” reverses the union/intersection quantifier. A small extra negative exponent converts uniform block control to Peter–Weyl square summability.

3 The counterexample

Theorem 1. *For every $1 \leq s < \infty$, the Dirac mass δ_e at the identity of $\mathrm{SU}(2)$ belongs to $(\gamma_{\mathbf{X}, L^2}^{(s)}(\mathrm{SU}(2)))'$, but it does not satisfy (3) for any $B > 0$. Consequently the printed second clause of Conjecture 4.5.5 is false.*

Proof. Source Definition 4.5.4 explicitly records

$$\mathcal{D}'(M) \subset (\gamma_{\mathbf{X}, L^2}^s)'(M) \subset (\gamma_{\mathbf{X}, L^2}^{(s)})'(M).$$

Thus δ_e belongs to the asserted Beurling dual. This also follows directly from Sobolev point evaluation on the compact three-manifold $\mathrm{SU}(2)$.

For every irreducible unitary representation π_l ,

$$\widehat{\delta}_e(l) = \pi_l(e)^* = I_{d_l}.$$

The smallest eigenvalue of $\widehat{\mathcal{L}}(l)$ is attained at $m = \pm l$ and equals l . Hence, for $l > 0$,

$$\begin{aligned} \|e^{B\widehat{\mathcal{L}}(l)^a} \widehat{\delta}_e(l)\|_{\mathrm{HS}}^2 &= \sum_{m=-l}^l e^{2B(l(l+1)-m^2)^a} \\ &\geq 2e^{2Bl^a}. \end{aligned}$$

For every $B > 0$ this tends to infinity with l , contradicting (3). □

The failure is not marginal: every finite linear combination of derivatives of δ_e has at most polynomially growing Fourier blocks and is still excluded by the positive exponential condition unless it vanishes.

4 The corrected theorem

We now make the natural topology explicit. For $D > 0$, let

$$H_D = \mathrm{Dom}(e^{D\mathcal{L}^a}), \quad \|f\|_D = \|e^{D\mathcal{L}^a} f\|_{L^2}.$$

The source’s $\mathrm{SU}(2)$ characterization identifies the Roumieu space with the inductive union $\bigcup_{D>0} H_D$. Its stated Beurling analogue is the projective intersection $\bigcap_{D>0} H_D$. We equip these with their standard inductive- and projective-limit locally convex topologies.

Theorem 2 (Exact sign correction). *Under these natural topologies:*

1. $u \in (\gamma_{\mathbf{X}, L^2}^s(\mathrm{SU}(2)))'$ if and only if (2) holds for every $B > 0$.

2. $v \in (\gamma_{\mathbf{X}, L^2}^{(s)}(\mathrm{SU}(2)))'$ if and only if there exist $B, K > 0$ such that

$$\|e^{-B\widehat{\mathcal{L}}(l)^a}\widehat{v}(l)\|_{\mathrm{HS}} \leq K \quad \text{for every } l \in \tfrac{1}{2}\mathbb{N}_0. \quad (4)$$

Thus the first printed clause is correct, and the second becomes correct after changing $+B$ to $-B$.

Proof. Duality of a directed Hilbert scale gives

$$\left(\bigcup_{D>0} H_D\right)' = \bigcap_{D>0} H_{-D}, \quad \left(\bigcap_{D>0} H_D\right)' = \bigcup_{D>0} H_{-D}. \quad (5)$$

For completeness, the first identity follows because a functional on the inductive limit is continuous on every H_D . For the second, continuity is bounded by finitely many defining seminorms, hence by one of the directed norms $\|\cdot\|_D$. Riesz representation on H_D then identifies the corresponding dual with H_{-D} .

Peter–Weyl and the spectral calculus give

$$\|w\|_{-D}^2 = \sum_{l \in \frac{1}{2}\mathbb{N}_0} d_l \|e^{-D\widehat{\mathcal{L}}(l)^a}\widehat{w}(l)\|_{\mathrm{HS}}^2. \quad (6)$$

Thus $w \in H_{-D}$ immediately implies a uniform block bound with exponent D .

Conversely, suppose the block bound holds with exponent B . If $D > B$, then (1) and $\min_m(l(l+1) - m^2) = l$ for $l > 0$ yield

$$\|w\|_{-D}^2 \leq C_0 + K^2 \sum_{l>0} (2l+1)e^{-2(D-B)l^a} < \infty.$$

The last series converges because $a > 0$; stretched-exponential decay dominates the polynomial factor. Therefore a uniform bound for every $B > 0$ is equivalent to membership in every H_{-D} , while a bound for some $B > 0$ is equivalent to membership in some H_{-D} . Combine this with (5). $\square$

5 Verification, novelty, and scope

An exact-arithmetic checker verifies the block spectrum and extremal eigenvalues for 50,000 half-integer representations, then stress-tests the divergent lower bound and the corrected-weight tail over a grid of s, B, D . These checks guard against indexing and sign mistakes; the proof above establishes the infinite statements.

Four run indexes and bounded exact-title/core-keyword searches found no later resolution through 11 August 2026. The argument fully refutes the printed Beurling clause and repairs it. It does not address the distinct Heisenberg conjecture, whose criterion is an L^1 -integrability statement rather than a uniform compact-group block bound. Human review should focus on the locally convex topology intended by the source for the strengthened corrected theorem; Theorem 1 itself is topology-robust because the source explicitly includes $\mathcal{D}'(\mathrm{SU}(2))$.

References

- [1] C. A. Taranto, *Wave equations on graded groups and hypoelliptic Gevrey spaces*, PhD thesis, Imperial College London; arXiv:1804.03544 (2018), Corollary 4.3.6 and Section 4.5.1.